

SAJJA PREM SAI

+91 6305078245 sajjapremsai.in@gmail.com Bengaluru, India github.com/SajjaPrem Sai sajjapremsai.github.io

EDUCATION

Avanthi Institute of Engineering and Technology
B.Tech in Computer Science (AI & ML)
CGPA: 8.31 / 10

2021 – 2025
Vizianagaram, India

EXPERIENCE

American Express
Apprentice – Enterprise Technology Services

Jan 2026 – Present
Bengaluru, India

- Contributing to enterprise AI modernization initiatives by evaluating and integrating GenAI frameworks, internal AI tooling, and agentic workflows using **LangChain** and **LangGraph** for engineering productivity applications.
- Developing MCP-based integrations to aggregate operational metrics including vulnerabilities, secrets, certificates, container insights, and repository analytics from enterprise platforms such as **GitHub Enterprise**, **DotConnector**, and **IRIS** systems.
- Building conversational application insights solutions that enable teams and leadership stakeholders to retrieve engineering and operational analytics through natural language interactions and AI-assisted tooling.
- Designing interactive **React.js** dashboards, centralized reporting systems, and operational visibility platforms for project health, security posture, and team-level analytics.
- Collaborating with CCE Ops teams and GenAI governance stakeholders to automate repetitive operational processes, reduce manual tracking efforts, and improve visibility for VP/SVP-level stakeholders.

PROJECTS

Visual Product Search Engine — Python, TensorFlow, Qdrant

[GitHub](#)

Designed and developed a vector-based visual search engine capable of retrieving visually similar products using deep learning embeddings. Built a CNN-based feature extraction pipeline to generate high-dimensional image embeddings and integrated the Qdrant vector database for efficient similarity search and nearest-neighbor retrieval. Achieved **90% retrieval precision** for jewelry image queries across the evaluation dataset.

Result Analysis Portal — PHP, MySQL

[Live](#)

Developed a web-based academic analytics platform that automates the processing and analysis of university examination results. Implemented a backend pipeline to extract structured data from JNTUK PDF result sheets and store them in a relational database. Built dashboards and performance analytics modules for tracking academic trends across **1000+ student records**.

Drowsiness Detection System — Python, OpenCV, CNN

Built a real-time driver fatigue detection system using computer vision techniques. Implemented Eye Aspect Ratio (EAR) based blink detection to identify early signs of drowsiness and integrated a real-time alert mechanism to improve driving safety.

TECHNICAL SKILLS

Languages	Python, Java, JavaScript, TypeScript(Learning and Developing Applications)
AI / ML	TensorFlow, CNNs, NLP, Embeddings
LLM / AI Agents	LangChain, LangGraph, MCP Servers, Prompt Engineering, Agentic AI
Backend	FastAPI, Node.js, Express.js, REST APIs
Frontend	React.js
Databases	MySQL, Vector Databases (Faiss)
Tools	Git, Docker, Postman, Linux CLI, MCP Inspector

ACHIEVEMENTS

- 1st Place – College Project Expo 2024
- 5th Place – Brainovision International Data Science Hackathon 2023
- Solved **250+ LeetCode problems** with a maximum rating of **1595**
- Active participant in **Kaggle competitions** focused on Machine Learning and Deep Learning problems